```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv(
    "/data.txt",
    sep=r"\s+",
    engine="python"
)

df.head()
```

	time	V(n002)	I(V1)
0	0.000000	145.6878	-3.890075e-05
1	0.000002	145.5795	2.529091e-07
2	0.000006	145.3629	2.529450e-07
3	0.000014	144.9297	2.529746e-07
4	0.000030	144.0633	2.530751e-07

Next steps: New interactive sheet

```
time = df["time"].values
signal = df["I(V1)"].values
```

```
dt = np.diff(time)

print(dt.min())
print(dt.max())
```

```
8.313561465078667e-07
2.000000000010327e-05
```

```
Fs = 50000          # choose sampling frequency (Hz)
dt = 1/Fs

t_uniform = np.arange(time[0], time[-1], dt)

signal_uniform = np.interp(
    t_uniform,
    time,
    signal
)
```

```
time = t_uniform
signal = signal_uniform
```

```
signal = signal - np.mean(signal)
```

```
N = len(signal)

fft = np.fft.rfft(signal)

freq = np.fft.rfftfreq(N, d=1/Fs)

magnitude = np.abs(fft) * 2 / N
```

```
plt.figure(figsize=(12,5))

plt.plot(freq, magnitude)

plt.xlim(0,500)
plt.xticks(np.arange(0,501,50))

plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
```

```
plt.grid(True)

plt.show()
```

